

Problem

An alternating voltage is given by $v(t) = 55 \cos(5t + 45^\circ)$ V. Use phasors to find

$$10v(t) + 4 \frac{dv}{dt} - 2 \int_{-\infty}^t v(t) dt$$

Assume that the value of the integral is zero at $t = -\infty$.

Step-by-step solution

Step 1 of 2

Consider the following data:

Given an alternating voltage, $v(t) = 55 \cos(5t + 45^\circ)$ V.

Given function is, $f = 10v(t) + 4 \frac{dv}{dt} - 2 \int_{-\infty}^t v(t) dt$

Rewrite voltage as shown below:

$$\begin{aligned} v(t) &= 55 \cos(5t + 45^\circ) \text{ V} \\ &= 55 (\cos 5t \cos 45^\circ - \sin 5t \sin 45^\circ) \\ &= 55 \left(\frac{1}{\sqrt{2}} \cos 5t - \frac{1}{\sqrt{2}} \sin 5t \right) \text{ V} \end{aligned}$$

Step 2 of 2

Evaluate the given function.

$$\begin{aligned}
 f &= 10v(t) + 4 \frac{dv}{dt} - 2 \int_{-\infty}^t v(t) \\
 &= \left[(10) [55 \cos(5t + 45^\circ)] + 4 \left(- (55) \left(\frac{1}{\sqrt{2}} \sin 5t \right) (5) - (55) \left(\frac{1}{\sqrt{2}} \cos 5t \right) (5) \right) \right] \\
 &\quad - 2 \int_{-\infty}^t \frac{55}{\sqrt{2}} (\cos 5t - \sin 5t) dt \\
 &= \left[550 \cos(5t + 45^\circ) - \frac{1100}{\sqrt{2}} (\sin 5t + \cos 5t) \right] \\
 &\quad - \frac{110}{\sqrt{2}} \left(\frac{\sin 5t}{5} + \frac{\cos 5t}{5} \right) \Big|_{-\infty}^t \\
 &= \left[550 \cos(5t + 45^\circ) - 1100 \left((\sin 5t) \left(\frac{1}{\sqrt{2}} \right) + (\cos 5t) \left(\frac{1}{\sqrt{2}} \right) \right) \right] \\
 &\quad - \frac{110}{5} \left((\sin 5t) \left(\frac{1}{\sqrt{2}} \right) + (\cos 5t) \left(\frac{1}{\sqrt{2}} \right) \right) \\
 &= \left[550 \cos(5t + 45^\circ) - 1122 \left((\sin 5t) \left(\frac{1}{\sqrt{2}} \right) + (\cos 5t) \left(\frac{1}{\sqrt{2}} \right) \right) \right] \\
 &= [550 \cos(5t + 45^\circ) - 1122 \sin(5t + 45^\circ)] \\
 &= [550 \cos(5t + 45^\circ) - 1122 \cos(5t + 45^\circ - 90^\circ)] \\
 &= [550 \cos(5t + 45^\circ) - 1122 \cos(5t - 45^\circ)] \\
 &= (550 \angle 45^\circ - 1122 \angle -45^\circ) \\
 &= 1249.5 \angle 108.9^\circ
 \end{aligned}$$

Therefore, the function value f is $\boxed{1249.5 \angle 108.9^\circ}$.